

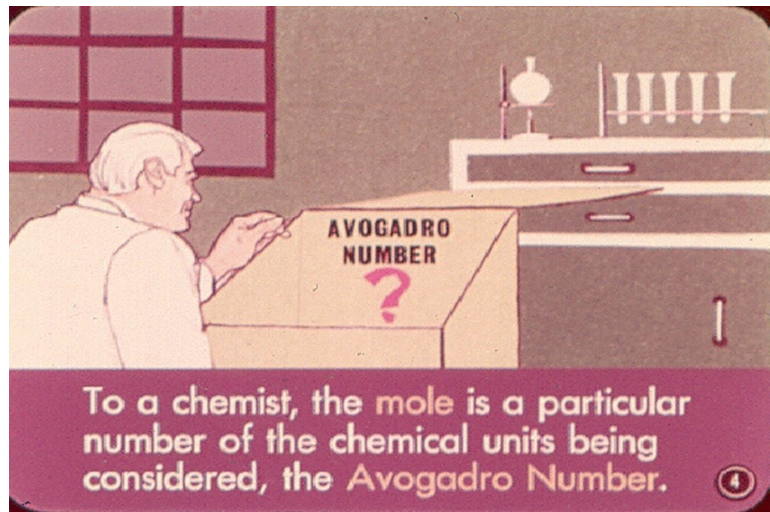
The Mole Concept

The Mole



$$6.022 \times 10^{23}$$

If you want to read more about this topic, read the e-book [section 7.4, 61, and 6.2](#) or the MEB book [unit 1](#)



To a chemist, the mole is a particular number of the chemical units being considered, the Avogadro Number.

Specific numbers of objects can be given names:



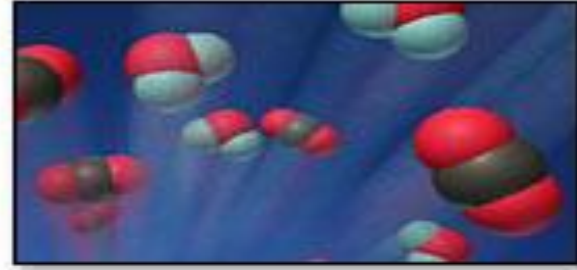
dozen

12



gross

144

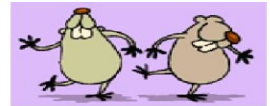


★ ★
★ mole ★

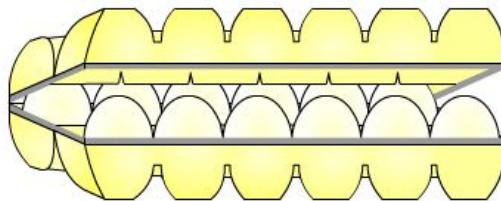
6.02×10^{23}

Also called Avogadro's
number

$$N_A = 6.02 \times 10^{23} = 1 \text{ mole of particles}$$



A **mole** is equal to 6.022×10^{23} items, in the same manner that a dozen equals 12 items.



$$1 \text{ mol } \cancel{\text{eggs}} \frac{6.022 \times 10^{23} \text{ eggs}}{\cancel{\text{mol}}} = 6.022 \times 10^{23} \text{ eggs} \quad 1 \text{ dozen } \cancel{\text{molecules}} \frac{12 \text{ molecules}}{\cancel{\text{dozen}}} = 12 \text{ molecules}$$

Avogadro's number is equal to: $6.022 \times 10^{23} \text{ items/mol}$

While you can have a mole of any item (like eggs), the most common units expressed per mole in chemistry are **molecules** and **atoms**.

Why 6.02×10^{23} ?

The mole is defined as the number of atoms in 12 grams of pure carbon-12.

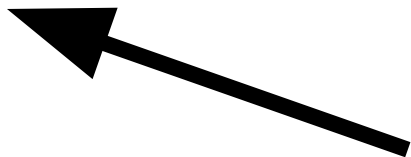
Atomic Mass Unit

$$1 \text{ amu} = 1/N_A \text{ g}$$

$$1 \text{ amu} = 1.660 \times 10^{-24} \text{ g}$$

Allows for
convenient
conversions
between atomic
mass units and
grams!!

The mass of a
single proton or
neutron.



Molar Mass

Mass of a substance that contains 1 mole of particles. Molar mass of an element is the mass of a sample of that element that contains 1 mole of its atoms. In other words; 6.02×10^{23} of its atoms.

Atomic mass (amu)	1 H 1.0079	6 C 12.011	92 U 238.029
	1 mole of naturally occurring H atoms has a mass of 1.0079 g.	1 mole of naturally occurring C atoms has a mass of 12.011 g.	1 mole of naturally occurring U atoms has a mass of 238.029 g.

Review Question: Why is the mass of Carbon not 12 amu exactly?

MOLAR MASS and its many names:

The mass of 1 mole of an atom or molecule is called **molar mass** and the unit is **g/mol**.

If the particle is an atom, we can also call it '**atomic mass**.'

We can read the atomic mass directly from the periodic table.

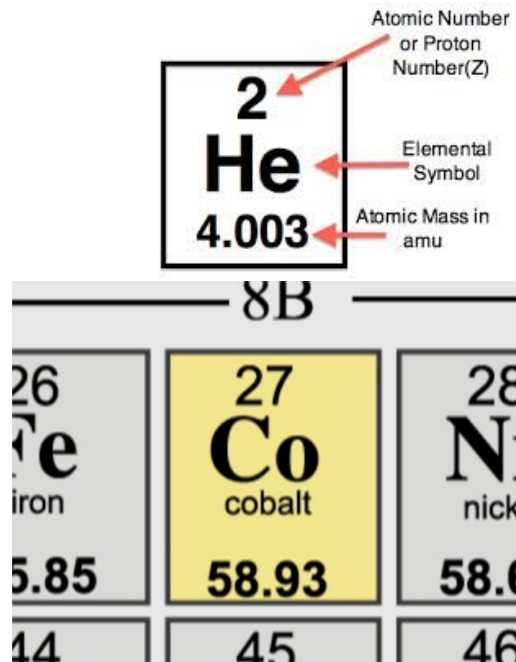


Diagram illustrating the periodic table structure and atomic data for Helium (He), Cobalt (Co), and Nickel (Ni). The diagram shows the periodic table grid with Helium (He) at the top, and Cobalt (Co) and Nickel (Ni) below it. Helium's box is highlighted with a red border and red arrows pointing to its atomic number (2), elemental symbol (He), and atomic mass (4.003). Cobalt's box is highlighted in yellow. Nickel's box is highlighted in grey. The atomic number (Z) is shown above the element symbol, the elemental symbol is in the center, and the atomic mass is below the symbol. The atomic number is also shown to the left of the element symbol. The atomic mass is shown below the element symbol. The atomic number is also shown to the left of the element symbol. The atomic mass is shown below the element symbol.

Element	Atomic Number (Z)	Elemental Symbol	Atomic Mass (amu)
Helium	2	He	4.003
Cobalt	27	Co	58.93
Nickel	28	Ni	58.6

If the particle is a molecule, we can call its molar mass '**molecular mass**' or '**molecular weight**' instead.

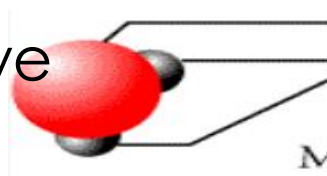


Diagram illustrating the molar mass calculation for water (H₂O). The diagram shows a water molecule (H₂O) with one red oxygen atom and two white hydrogen atoms. Lines connect the atoms to their respective molar masses: Oxygen (O) is 15.9994 g/mol, and Hydrogen (H) is 1.00794 g/mol. The total molar mass for H₂O is calculated as 18.0153 g/mol.

Component	Molar Mass (g/mol)
Molar mass O:	15.9994
Molar mass H:	1.00794
Molar mass H ₂ O:	18.0153

$$\begin{aligned}
 &1 \text{ mole of hydrogen atom} = 1 \text{ g} \\
 &\text{or } 6.022 \times 10^{23} \text{ atoms of hydrogen weigh} = 1 \text{ g} \\
 &\text{Mass of one atom} = \frac{1}{6.022 \times 10^{23}} \text{ g} \\
 &= 1.66058 \times 10^{-24} \text{ g}
 \end{aligned}$$

Molar mass of Ag is 108.0 g/mol. What is the mass of a single Ag atom?

93	Copper 63.546	Zinc 65.38
42	47 Ag Silver 107.87	48 Cd Cadmium 112.41
	79 Au	80 Hg

Mole amount can be formulated as

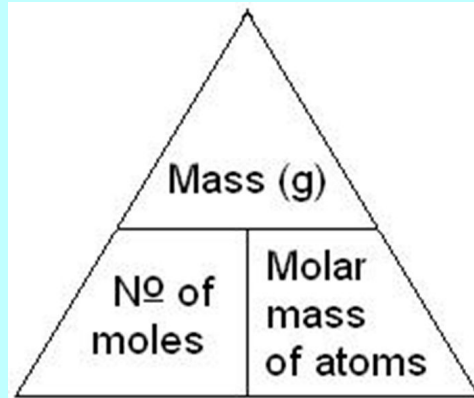
$$n = \frac{\text{Number}_{\text{GIVEN}}}{N_A}$$

Number of
moles
(mole
amount)

Number of
particles you
have

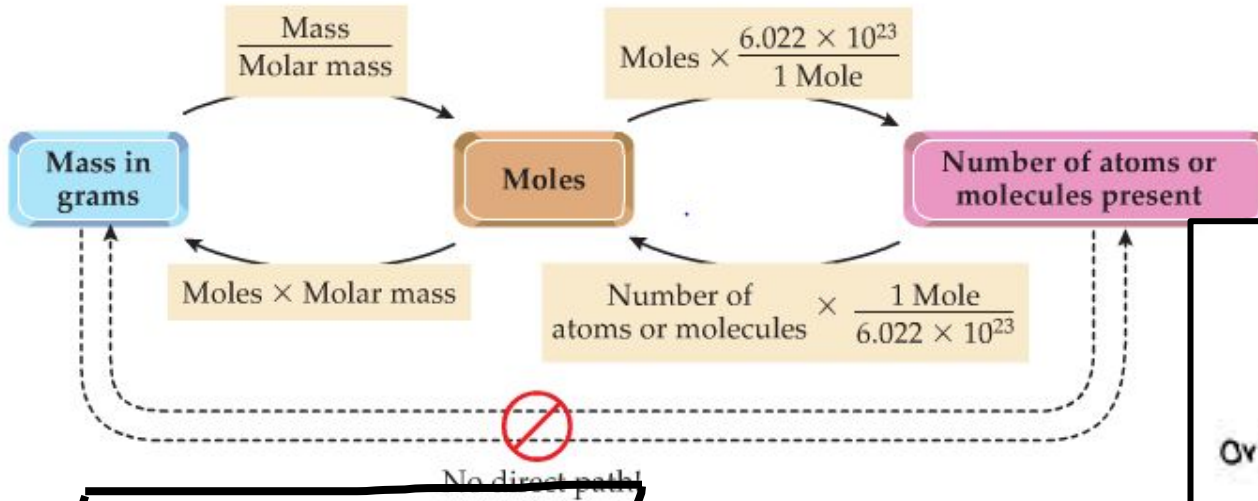
Avogadro's
Number

Mole Triangle



Converting between moles and grams

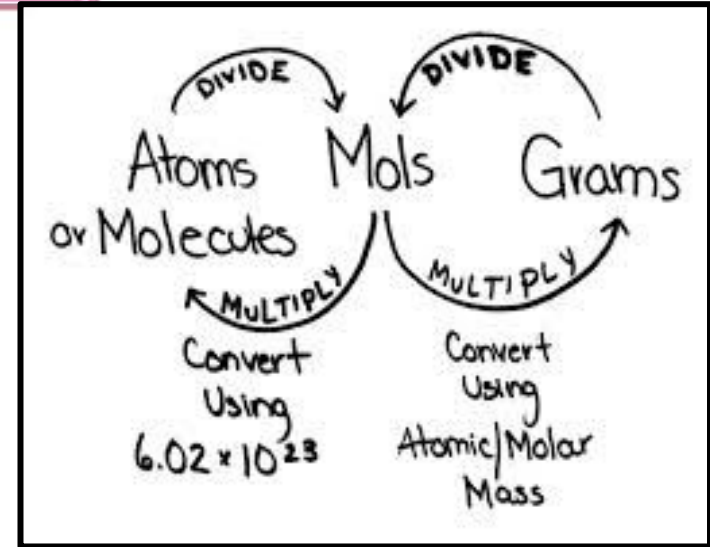
Conversions in Stoichiometry Calculations



No direct path!

$$n = \frac{m}{M_m}$$

n = number of moles
 m = mass
 M_m = molar mass



Ex: How many moles do 12.04×10^{22} water molecules make?

1 mol water molecule $\rightarrow 6.02 \times 10^{23}$ H_2O molecules

? mol water molecule $\rightarrow 12.04 \times 10^{22}$ H_2O molecules

$$n = \frac{1 \times 12.04 \times 10^{22}}{6.02 \times 10^{23}} = 2 \times 10^{22-23} = 2 \times 10^{-1} = 0.2 \text{ mol}$$

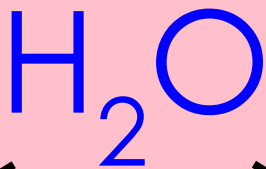
Ex: How many molecules are there in 0.015 mol CO₂ gas?

1 mol CO_2 molecule $\rightarrow 6.02 \times 10^{23}$ CO_2 molecules

0.015 mol CO_2 molecule $\rightarrow x = ?$ CO_2 molecules

$$n = 0.015 \times 6.02 \times 10^{23} = 0.09 \times 10^{23} = 9.03 \times 10^{21} \text{ molecules}$$

In a single molecule



2 H atoms

1 O atom

How many moles of H and O atoms are there in 1 *mole* of water molecule?

1 mol of H_2O molecule contains mol of H atoms and mol of O atoms.

1 mol H_2O molecule contains 2 mol of H atoms and 1 mol of O atoms.

$$2 \text{ mol} \times 6.02 \times 10^{23} = 12.04 \times 10^{23} \quad \text{H atoms}$$

$$1 \text{ mol} \times 6.02 \times 10^{23} = 6.02 \times 10^{23} \quad \text{O atoms}$$

Question 1:

How many atoms are there in 0.042 mol of Fe atoms?

Question 2:

How many moles of Na sample contain 3.01×10^{21} Na atoms?

Question 3:

How many molecules do 0.023 moles of H_2O_2 molecules contain?

Question 4:

Calculate the amount of moles of 8.32×10^{24} molecules of MgO .

Question 5:

0.04 mol Fe_2O_3 contains:

- a) how many mol of Fe atoms and O atoms?
- b) how many Fe atoms and O atoms?
- c) how many atoms totally?
- d) how many molecules?

Question 6:

..... mol $\text{C}_2\text{H}_5\text{OH}$ contains 0.12 mol H atoms and mol C atoms.

Some examples...

Ex: If 1000 atoms of an element, X, weighs $9.30 \times 10^{-20}\text{g}$, find the molar mass (atomic mass) of element of X.

Ex: Calculate the molar mass (molecular mass) of $\text{Ca}_3(\text{PO}_4)_2$:
(Ca=40.0, P=31.0, O=16.0)

Ex: Calculate the molar mass of iron (III) nitrate.
Find the atomic masses that you need from the periodic table.

Ex: 0.2 mol Fe_2O_3 containsg Fe andg O.

Ex: Find the masses of the following samples.

a) 0.300 moles of oxygen gas

b) 1.62 moles of CaCO_3

Ex: Make the following conversions.

5.00 g Ca = mol of Ca atoms.

0.005 mol O atoms =g.

0.980 g H_2SO_4 = mol.

0.06 mol NaCl = g.

Ex: 3 mol H_2SO_4 containsg H atoms,
.....g S atoms, and g O atoms.

Ex: How many oxygen atoms does 6.40 g SO_2 gas sample contain?

Ex: 6.4g SO_2 containsg S atoms
and..... g O atoms.

Determining Atomic and Molecular Mass

Ex: 3.42 g $X_2(SO_4)_3$ contains 0.12 mol oxygen atoms. What is the atomic mass of X atom? (Answer: 27 g/mol)

Ex: 0.25 mol of C_nH_{2n-2} contains 1.50 mol of H atoms. What is the molecular mass of the compound? (Answer: 54 g/mol)

Ex: 1.88 g of X_2O contains 1.56 g of X.

a) Calculate the molecular mass of this compound.

b) Calculate the atomic mass of the element X.

(Answers: 94 g/mol and 39 g/mol)

MOLAR VOLUME

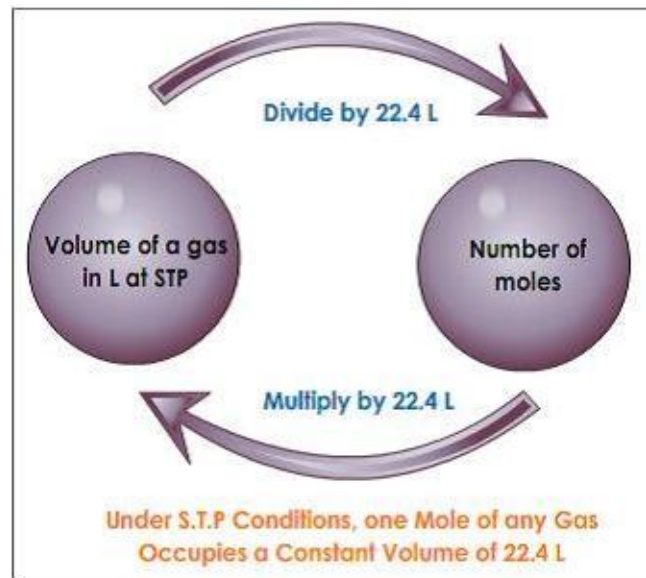
Represents the volume of 1 mole of a sample.

At STP (Standard Temperature and Pressure); 1 mole of any gas occupies a volume of 22.4 L.

STP conditions are: 0 °C or 273K and 1 atm

Ex 1: Find the mole number of a HCl gas sample that occupies a volume of 44.8 L at STP.

Ex 2: How many grams does a hydrogen gas sample weigh that has 6.72 L of volume at STP?



Ex 3: How many molecules of CO_2 gas occupies a volume of 2.24 L at STP?



Ex 4: What is the volume of a SO_3 gas sample at STP; that contains 9.63×10^{24} atoms?

Percent Composition By Mass

$$\% \text{ by mass} = \frac{\text{Total Molar Mass of the element in the compound}}{\text{Molar mass of the compound}} \times 100$$

Ex: Calculate the percent composition by mass of oxygen in the compound $\text{Al}_2(\text{SO}_4)_3$

Empirical and Molecular Formulae

The Empirical formula is the simplest formula of a compound, it shows the simplest ratio between the elements.

The Molecular formula is the actual formula of a compound, that gives the actual numbers of atoms of each element in the compound.

Ex: Molecular Formula of Glucose is $C_6H_{12}O_6$; which tells us that a single glucose molecule contains 6 C, 12 H and 6 O atoms.
Empirical Formula of this compound is CH_2O ; which tells us that the combining ratio of C:H:O is 1:2:1 in glucose.

Ex: Ethanol is composed of 52.2 % C, 13.0 % H and 34.8 % O atoms by mass.

a) What is the empirical formula of this compound? (C=12.0, H=1.01, O=16.0)

b) If the molar mass is 46 g/mol, what is the molecular formula?

Ex: An organic compound is composed of C, H and O elements.

8.40 g of this compound contains 2.4×10^{23} C atoms and 0.40 g of H atoms.

If this sample has $N_A/10$ molecules, what is the molecular formula of the compound? Answer: $C_4H_4O_2$